

Normal Gaussian log likelihood's solution

1. let $X = [x_1, x_2, x_3 \dots x_n]^T$, $x_1, x_2 \dots x_n$ is iid.

$$p(X|\mu, \sigma^2) = \prod_{n=1}^N \mathcal{N}(x_n|\mu, \sigma^2)$$

$$= \prod_{n=1}^N \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^{1/2} \cdot \exp\left[-\frac{1}{2\sigma^2}(x_n - \mu)^2\right]$$

2. take log.

$$\ln p(X|\mu, \sigma^2) = \sum_{n=1}^N \left[-\frac{1}{2\sigma^2}(x_n - \mu)^2 - \frac{1}{2}\ln 2\pi - \frac{1}{2}\ln \sigma^2\right]$$

$$t) \mu_{ML} \left(\frac{\partial}{\partial \mu} \ln p = 0\right) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2} \ln 2\pi - \frac{N}{2} \ln \sigma^2$$

$$\frac{\partial}{\partial \mu} \ln p(X|\mu, \sigma^2) = -\frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu)(-1) = \frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu) = 0$$

$$\Rightarrow \sum_{n=1}^N x_n = \sum_{n=1}^N \mu \Rightarrow \sum_{n=1}^N x_n = N \cdot \mu$$

$$\Rightarrow \mu = \frac{1}{N} \sum_{n=1}^N x_n = \mu_{ML} \text{ (sample mean)}$$

$$2) \sigma_{ML}^2 \left(\frac{\partial}{\partial \sigma^2} \ln p = 0\right)$$

$$\frac{\partial}{\partial \sigma^2} \ln p(X|\mu, \sigma^2) = \frac{1}{2(\sigma^2)^2} \sum_{n=1}^N (x_n - \mu)^2 - \frac{N}{2\sigma^2} = 0$$

$$\Rightarrow \sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2$$

(sample variance)

(combine both result gives

$$\sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2$$

(sample variance with respect to the sample mean μ_{ML})

Expectation of μ_{ml}, σ_{ml}^2 (x is identically distributed)

$$E[\mu_{ml}] = E\left[\frac{1}{N} \sum_{n=1}^N x_n\right] = \frac{1}{N} \sum_{n=1}^N E[x_n] \quad (\text{linearity})$$

$$= \frac{1}{N} \sum_{n=1}^N (\mu) = \mu$$

$$E[\sigma_{ml}^2] = E\left[\frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ml})^2\right] = \frac{1}{N} \sum_{n=1}^N E\left[x_n^2 - 2\mu_{ml}x_n + \mu_{ml}^2\right]$$

$$= \frac{1}{N} \sum_{n=1}^N \left(E[x_n^2] - 2E[\mu_{ml}x_n] + E[\mu_{ml}^2]\right) \quad (\text{linearity})$$

$$= \frac{1}{N} \sum_{n=1}^N \left(E[x_n^2] - \underbrace{E[\mu_{ml}^2]}_{?}\right)$$

$$(\because E[\mu_{ml}^2] = E\left[\mu_{ml} \frac{1}{N} \sum_{n=1}^N x_n\right] = \frac{1}{N} \sum_{n=1}^N E[\mu_{ml}x_n] = E[\mu_{ml}x_n],$$

$$\therefore E[\mu_{ml}x_1] = E[\mu_{ml}x_2] = \dots = E[\mu_{ml}x_n])$$

$$E[\mu_{ml}^2] = E\left[\left(\frac{1}{N} \sum_{n=1}^N x_n\right)^2\right] = \frac{1}{N^2} \sum_{n=1}^N E\left[\left(\sum_{m=1}^N x_m\right)x_n\right] \quad (\text{linearity})$$

$$= \frac{1}{N^2} \sum_{n=1}^N \left(E[x_n^2] + E\left[\sum_{\substack{m=1 \\ m \neq n}}^N x_m x_n\right]\right) \quad \text{where } m \neq n$$

$$= \frac{1}{N^2} \sum_{n=1}^N \left(\mu^2 + \sigma^2 + E\left[\sum_{m=1}^N x_m\right] \cdot E[x_n]\right) \quad \begin{matrix} \text{for using uncorrelated} \\ \text{assumption} \end{matrix} \quad (\text{uncorrelated assumption})$$

$$= \frac{1}{N^2} \sum_{n=1}^N \left(\mu^2 + \sigma^2 + (N-1)\mu^2\right) = \mu^2 + \frac{\sigma^2}{N}$$

$$\therefore E[\sigma_{ml}^2] = \frac{1}{N} \sum_{n=1}^N \left(\mu^2 + \sigma^2 - \mu^2 - \frac{\sigma^2}{N}\right) = \frac{N-1}{N} \sigma^2$$